JACKSON MORRIS

CURRICULUM VITAE

Simon Laufer Mathematical Science Institute
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EMPLOYMENT

Postdoctoral Researcher — *January 2027-August 2029*
Vanderbilt University

Postdoc SLMath — *August-December 2026*
SLMath Semester Program
Motivic Homotopy Theory: Connections and Applications

EDUCATION

Ph.D. in Mathematics — *September 2021-June 2026*
University of Washington
Advisors: Kyle Ormsby and John Palmieri
Thesis title: Cooperations in motivic homotopy theory

M.S. in Mathematics — *September 2021-June 2023*
University of Washington

B.S. in Mathematics, *magna cum laude* — *September 2017-December 2020*
University of Kentucky

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

9. *Diachromatic blueshift*, joint with Kyle Ormsby. In preparation.
8. *Motivic K-theory cooperations over the rationals,* joint with Sarah Petersen and Liz Tatum. In preparation.
7. *An exotic self-map of periodicity 1.* In preparation.
6. *Periodic phenomena in stable motivic homotopy theory.* Submitted. Available at https://arxiv.org/abs/2607.18165
5. *Immersions of C_2-projective spaces via* $\mathrm{K}\mathbb{R}$*-theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Submitted. Available at https://arxiv.org/abs/2604.25260.
4. *Spectrum level splittings of truncated motivic Brown–Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at https://arxiv.org/abs/2509.19542.
3. *Rings of cooperations for hermitian K-theory over finite fields.* To appear in **Algebraic & Geometric Topology**. Available at https://arxiv.org/abs/2509.02786.
2. *On the ring of cooperations for real hermitian K-theory.* To appear in **Annals of K-theory**. Available at https://arxiv.org/abs/2506.16672.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra 49** (2020), **7**, 3085-3093.

SELECTED PRESENTATIONS & SEMINAR TALKS

Conference and Colloqium Talks

"Higher Witt K-theories"
AMS Sectional: Computational Homotopy Theory (Tempe, AZ) *November 2026*
Young Topologists Meeting 2026 (Copenhagen, DK) *July 2026*
Cascade Topology Seminar (Portland, OR) *May 2026*

"Splittings and cooperations in motivic homotopy theory"
Joint Math Meetings 2026 (Washington, D.C.) *January 2026*

"On the ring of cooperations for real hermitian K-theory"
European Talbot 2025 (Kolding, DK) *July 2025*
Graduate Student Geometry & Topology Conference 2025 (Bloomington, IN) *April 2025*
Joint Math Meetings 2025 (Seattle, WA) *January 2025*
European Autumn School in Topology (Utrecht, NL) *September 2024*
Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA) *August 2024*

"Finding Fixed Points with Invariants"
Reed College Mathematics Colloquium *February 2025*

Seminar Talks

eCHT Research Seminar *September 2026*
University of Washington Topology Seminar *February 2026*
University of Illinois Urbana-Champaign Topology Seminar *January 2026*
University of Virginia Topology Seminar *November 2025*
University of Kentucky Topology Seminar *November 2025*
Duke University Geometry and Topology Seminar *October 2025*
University of Colorado Boulder Homotopy Theory Seminar *September 2025*
University of Michigan Geometry Seminar *September 2025*
University of Notre Dame Topology Seminar *September 2025*
University of Washington Topology Seminar *June 2025*

Contributed Talks

eCHT Reading Seminars
"An upper bound on the degree of symmetry of an exotic sphere" *March 2026*
"Higher Adams differentials and hidden extensions" *November 2025*
"Motivic twisted K-theory" *April 2024*

Graduate Student Seminars
"Higher real K-theories, redshift, and blueshift" *May 2026*
"Filtered objects, spectral sequences, and t-structures" *January 2026*
"(A synthetic approach to detecting) v_1-periodic families" *November 2025*
"Algebraic and Milnor K-theory" *October 2025*
"Galois descent and the Picard group of K-theory" *May 2025*
"The Adams spectral sequence and Hopf algebroids" *February 2025*
"Symplectic orientations' *August 2024'*

Workshops
"$\pi_*\mathrm{THH}(\mathbb{Z}_p^{\mathrm{fil}})/p$ and $\pi_*\mathrm{THH}(\ell_p^{\mathrm{fil}})/(p, v_1)$" (Talbot, Nacogdoches TX) *August 2024*
"Hochschild Homology" (European Autumn School in Topology, Utrecht, NL) *September 2024*

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology *2025-Present*
ZBMath *2026-Present*

Seminar Organization

DUBTOP Seminar *2025-2026*
Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus

Back-To-School seminar *Fall 2024*
Student lectures on fun math learned over the summer.

Reading seminars *2023-2024*
Topics included: Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories

TEACHING EXPERIENCE

Instructor of Record

University of Washington
Math 480 – Homotopy theory *Spring 2026*
Math 208 – Linear algebra *Winter 2023, Summer 2023*

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)
Research Seminar *2025-2026*
Quadratic Curve Counting *Fall 2024*

Park City Math Institute (PCMI) *Summer 2024*
Experimental Math Lab

University of Washington *Fall 2021 - Spring 2026*
Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Directed Reading Mentor

University of Washington *2022-2025*
Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra

Math Olympiad Judge

UW Math Hour *2023 - 2024*
Judge for elementary school math competition

HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship *Summer 2025*
University of Washington *$750*
For travel to European Talbot in Kolding, DK.

Research Assistantship *Summer 2025*
University of Washington
For good research progress.

JC Eaves Scholarship *2020*
University of Kentucky *$2000*
For continued excellence in mathematics as a Kentuckian.

Carolyn Bunyan Award *2020*
University of Kentucky *$2000*
For mathematics majors interested in continuing education in mathematics.

SELECTED WORKSHOP AND CONFERENCE PARTICIPATION

Extended Research Visits

SLMath Semester Program: Motivic Homotopy Theory (Berkeley, CA) *Fall 2026*
PCMI: Motivic Homotopy Theory (Park City, UT) *Summer 2024*

Conference Participation

AMS Sectional: Computational Homotopy Theory (Tempe, AZ) *November 2026*
Algebraic Structures in Topology (San Juan, PR) *July 2026*
Young Topologists Meeting (Copenhagen, DK) *June/July 2026*
Cascade Topology Seminar (Portland, OR) *May 2026*
(∞,∞)**-Categories Workshop** (Chicago, IL) *March 2026*
JMM 2026 (Washington, D.C.) *January 2026*
Midwest Topology Seminar (Minneapolis, MN) *October 2025*
European Talbot (Kolding, DK) *July 2025*
WAGS Spring 2025 (Vancouver, CAN) *April 2025*
GSTGC 2025 (Bloomington, IN) *April 2025*
JMM 2025 (Seattle, WA) *January 2025*
SLMath Hot Topics: Life after the Telescope Conjecture (Berkeley, CA) *December 2024*
European Autumn School in Topology (Utrecht, NL) *September 2024*
Talbot 2024 (Nacogdoches, TX) *August 2024*
Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA) *August 2024*
eCHT Research Workshop on Hopf Rings (Online) *June 2024*
JMM 2024 (San Francisco, CA) *January 2024*
Scissors Congruence, Algebraic K-theory, & Trace Methods (Bloomington, IN) *July 2023*
Cascade Topology Seminar (Vancouver, CAN) *April 2023*
PRIMA Congress (Vancouver, CAN) *December 2022*
Graduate Students in Geometry, Topology, and Algebra (Philadelphia, PA) *May 2022*

REFERENCES

Mark Behrens mbehren1@nd.edu
John and Margaret McAndrews Professor of Mathematics
University of Notre Dame
Relationship: Research mentor

Kyle Ormsby ormsbyk@reed.edu
Mathematics Department Chair
Reed College
Relationship: PhD advisor and research collaborator

John Palmieri jpalmier@uw.edu
Professor of Mathematics
University of Washington
Relationship: PhD advisor; Mathematics Department Chair during first 2 years of graduate study

J.D. Quigley mbp6pj@virginia.edu
Assistant Professor of Mathematics
University of Virginia
Relationship: Research mentor; Program supervisor and mentor at PCMI

Charles Camacho camachoc@uw.edu
Assistant Teaching Professor
University of Washington
Relationship: Teaching supervisor and mentor

Natalie Naehrig naehrn@uw.edu
Assistant Teaching Professor
University of Washington
Relationship: Teaching supervisor and mentor